Hypergraph Representation Learning for Cancer Drug Response Prediction

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Abstract. Accurately predicting drug response is crucial for personalized cancer treatment. Current graph neural network methods mainly focus on pairwise relationships between cell lines and drugs, neglecting higher-order interactions. In this work, we propose a novel computational method called HRLCDR based on Hypergraph representation learning for predicting cancer drug responses. Firstly, HRLCDR constructs hypergraphs for cell lines and drugs based on their similarities, respectively. Subsequently, HRLCDR applies low-pass and high-pass hypergraph convolutions on these hypergraphs to capture the common and different features from the complex high-order interactions between nodes. Then it builds a heterogeneous graph based on the known cell line responses to drugs and employs parallel heterogeneous graph convolutions to learn primary interaction features of drugs and cell lines from known cell line-drug associations. Finally, HRLCDR integrates the cell line features and drug features learned from the hypergraphs and the heterogeneous graph to reconstruct the relationships between cell lines and drugs. We tested our model on the Cancer Drug Sensitivity Data (GDSC) and the Cancer Cell Line Encyclopedia (CCLE) databases. The results indicate that HRLCDR exhibits superior performance in predicting cancer drug responses compared with other state-of-the-art methods. The source code is available at https://github.com/weiba/HRLCDR.

Keywords: Hypergraph representation learning · hypergraph convolution · drug sensitivity prediction

1 Introduction

Accurately predicting anti-cancer drug response is a crucial step in precision medicine, enabling the tailoring of personalized cancer treatments to individual patients based on their unique genetic makeup. Traditional approaches rely on time-consuming experiments and clinical trials to assess drug efficacy [1]. Advancements in high-throughput sequencing technologies, along with the emergence of preclinical anti-cancer drug screening projects such as the Cancer Cell Line Encyclopedia (CCLE) [2] and the

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Genomics of Drug Sensitivity in Cancer (GDSC) [3], have provided researchers with wealthy drug response data of cell lines, which empowers the construction of computational models to predict anti-cancer drug responses. These models serve as a powerful tool for improving the effectiveness and efficiency of cancer treatment, ultimately enhancing patient survival rates and reducing the costs of therapy.

There are many sophisticated computational models fusing cell line features and drug features to predict drug responses. According to the different ways of feature fusion, we divide current algorithms for predicting anti-cancer drug responses into two categories, machine learning-based method and network embedding-based method. Machine learning-based methods transform features of cell lines and drugs and fuse them into classifiers such as Support Vector Machines (SVM), Multilayer Perceptron (MLP) [4], Deep Forest [5], and Convolutional Neural Networks (CNN) [6] to predict drug responses. These methods usually adopt MLP [6], autoencoders (AE) [4], and Cascade Forest [5] to extract cell line features from gene expression profiles or multi-omics data and learn drug features from fingerprints by MLP [4] or from drug molecular graph by Graph Neural Networks (GNNs) [6]. After that, these methods concatenate the cell line features and drug features into a classifier to predict the cell line-drug reactions. However, these approaches mentioned above usually fail to consider the associations between drugs and cell lines comprehensively.

The network embedding-based method assume that similar cell lines have similar reactions with similar drugs. They usually construct heterogeneous networks with cell lines and drugs as nodes and the known responses as edges. These methods infer novel cell line responses to drugs by referring to neighbors in the network. They typically learn low-dimensional representations or embeddings of cell lines and drugs in the latent space while keeping the network's connection information and node attributes. Finally, they infer novel cell line responses to drugs by reconstructing the associations between cell line and drug using their latent features. Graph Neural Network (GNN) is a popular deep learning method based on heterogeneous graphs for predicting anti-cancer drug responses. The model can incorporate the node attributes with the network structure for more comprehensive node feature learning. For example, Peng et al. [7] propose MOFGCN, a model designed to predict drug response in cell lines by integrating multi-omics data of cell lines and drug molecular fingerprints. It implements a graph convolutional on a heterogeneous cell line-drug network to learn node features by aggregating the features of their neighbors in the network. After that, it infers drug responses from the reconstructed cell line-drug response matrix. GraphCDR [8] constructs two network views based on whether a cell line is sensitive or tolerant to specific drugs. Then, it introduces a contrastive learning framework into the graph convolutional aggregation module to enable the acquisition of higher-quality node features for cancer drug response predictions. NIHGCN [9] utilizes two separate graph convolutions to learn features for cell lines and drug simultaneously. Meanwhile, it introduces element-wise interactions between a node and its neighbors in the network to improve feature learning and anti-cancer drug response prediction. Besides the interactions between drugs and cell lines in the heterogeneous network, TSGCNN [10] introduces the drug similarity information and cell line similarity information in the homogeneous nodes. It proposes a dual-space graph convolutional neural network model to learn features for drug and cell line from both their heterogeneous and homogeneous neighbors. Ultimately, drug responses are predicted based on the reconstructed associations between cell lines and drugs using their dual-space node features. These network embedding-based methods usually learn node features from directly connected neighbors in the heterogeneous and homogeneous network, failing to consider the higher-order interaction information of nodes. The cell response information can be propagated along with the path in the heterogeneous and homogeneous network. Hence, it is necessary to design effective methods to learn features for cell lines and drugs from their high-order neighbors.

Hypergraphs represent a generalization of graphs wherein an edge can connect an arbitrary number of vertices with the same attributes. Besides the pair of nodes in the network, the hypergraph considers indirect high-order correlations and models more complex relationships. Hypergraphs have found extensive applications across fields to capture high-order correlations [11]. To address the issues above, we propose a computational method for predicting Cell line response to Drugs based on Hypergraph Representation Learning called HRLCDR. This method initially constructs hypergraphs of cell lines and drugs based on their similarities. Subsequently, it applies both low-pass and high-pass hypergraph convolutions on these hypergraphs to capture the common and different features from the high-order interactions between nodes. Then, HRLCDR builds a heterogeneous graph based on the known cell line responses to drugs and employs parallel heterogeneous graph convolutions to learn features for cell lines and drugs from known cell line-drug associations. Finally, it reconstructs the cell line-drug associative relationships by combining the features learned from the hypergraph and the features from heterogeneous graph. Experimental results indicate that our model performs excellently in predicting anti-cancer drug responses on the GDSC and CCLE datasets.

2 Materials

Our study utilized reaction data between cell lines and drugs from two sources: The Genomics of Drug Sensitivity in Cancer (GDSC) [3] and the Cancer Cell Line Encyclopedia (CCLE). From the GDSC database, we downloaded Tables S4A and S5C. Following established methods [9], a drug was considered sensitive for a cell line if its IC50 value in Table S4A exceeded a threshold defined in Table S5C. Conversely, drugs with IC50 values below the threshold were classified as resistant. We obtained data for 962 cancer cell lines responding to 265 drugs (20,851 sensitive samples and 156,512 resistant samples) after applying these GDSC preprocessing steps. For the CCLE dataset, we built an experimental benchmark using data on 11,670 drug assays from cell lines. We normalized IC50 values based on prior studies [10]. A cell line was sensitive to a drug if the normalized IC50 was below -0.8, and resistant otherwise. Following this preprocessing, we obtained response data for 24 drugs across 436 cell lines (1,696 sensitive samples and 8,768 resistant samples) from the CCLE dataset.

3 Methods

3.1 Overview

HRLCDR predicts cell line-drug responses in three steps (Fig. 1). First, it constructs separate hypergraphs for cell lines and drugs based on their similarities. Second, HRL-CDR employs low-pass and high-pass hypergraph convolution on these hypergraphs. Meanwhile, HRLCDR utilizes parallel heterogeneous graph convolution to learn node features from known cell line-drug associations. Finally, HRLCDR integrates the features learned from the high-pass and low-pass hypergraph convolutions, along with the features learned from the heterogeneous graph convolution. It leverages linear correlation coefficients to predict the strength of a cell line's response to a drug.

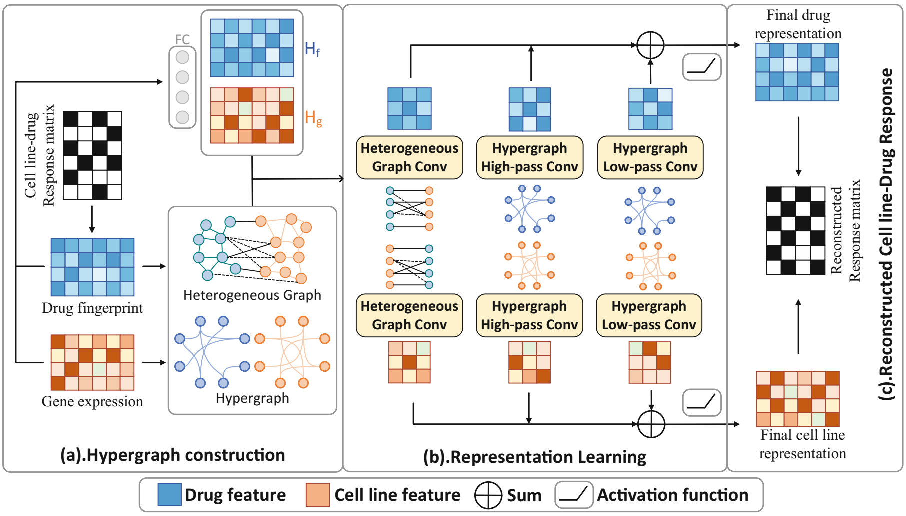

Fig. 1. The framework of HRLCDR

3.2 Hypergraph Construction

Based on the assumption that similar drugs have similar responses in similar cell lines, we constructed drug hypergraphs and cell line hypergraphs using similarity matrices for drugs and cell lines, along with a response matrix, to capture higher-order associations within drugs and cell lines [12].

Firstly, we calculate the drug similarity matrix S_d using Eq. 1:

$$S_{d(ij)} = \begin{cases} Sim_{d(ij)} & j \in K_i \\ 0 & j \notin K_i \end{cases} \tag{1}$$

where, $Sim_{d(ij)} = \frac{|d_i \cap d_j|}{|d_i \cup d_j|}$ denotes the Jaccard similarity between drug fingerprints, K_i represents the set of the top K drugs most similar to drug, K is set to 10.

Subsequently, we utilize Eq. 2 to calculate the matrix of cell line similarity:

$$S_{c(ij)} = \begin{cases} Sim_{c(ij)} & j \in T_i \\ 0 & j \notin T_i \end{cases} \tag{2}$$

where, $Sim_{c(ij)} = e^{-\frac{\|X_i - X_j\|_2}{2\varepsilon^2}}$ represents the exponential similarity of gene expression between cell lines, where X_i denotes the normalized gene expression values of the cell lines i, and T_i denotes the set of the top T cell lines most similar to the cell line i, with T also set to 10.

Let A denote the known cell line-drug response matrix. Inspired by [13], we calculated the hypergraph of drug H_d using the following Equations to capture the higher-order information of drug associations:

$$HD_1 = (S_d \cdot S_d) \odot S_d \tag{3}$$

$$HD_2 = \left(A^T \cdot A\right) \odot S_d \tag{4}$$

$$HD_3 = \left(A^T \cdot S_c \cdot A\right) \odot S_d + \left(\left(A^T \cdot S_c \cdot A\right) \odot S_d\right)^T \tag{5}$$

$$HD_4 = \left(S_d \cdot A^T\right) \cdot \left(S_d \cdot A^T\right)^T \tag{6}$$

$$H_d = HD_1 + HD_2 + HD_3 + HD_4 \tag{7}$$

where, HD_1 stores the drug similarities between similar drugs, HD_2 represents similar cell line response between similar drugs, HD_3 indicates an indirect similar cell line correlation between similar drugs, and HD_4 complements the direct correlation relationship of the drugs. Ultimately, a summation of these higher-order matrices yields the drug hypergraph, denoted as H_d.

Similarly, we computed the hypergraph of cell line H_c to capture the higher-order information of cell line associations by the following equations:

$$HC_1 = (S_c \cdot S_c) \odot S_c \tag{8}$$

$$HC_2 = \left(A \cdot A^T\right) \odot S_c \tag{9}$$

$$HC_3 = \left(A \cdot S_d \cdot A^T\right) \odot S_c + \left(\left(A \cdot S_d \cdot A^T\right) \odot S_c\right)^T \tag{10}$$

$$H_c = HC_1 + HC_2 + HC_3 \tag{11}$$

3.3 Initial Features of Cell Lines/Drugs

Initially, we employ Eq. 12 to normalize the expression levels of each gene within the cell lines, thereby obtaining $\overline{expr_i}$:

$$\overline{expr_i} = \frac{(expr_i - u_i)}{\sigma_i} \tag{12}$$

where, $expr_i$ denotes the expression values of the *ith* gene across all cell lines, with a mean of u_i and a standard deviation of σ_i.

Subsequently, we derive the cell line initial features $X_g \in R^{m \times d_g}$ via Eq. 13:

$$X_g = C_g \cdot W_g \tag{13}$$

where, $C_g \in \mathbb{R}^{m \times d_{cg}}$ represents the normalized gene expression values in cell lines obtained by Eq. 12, with m denoting the number of cell lines, d_{cg} the count of genes, and $W_g \in \mathbb{R}^{d_{cg} \times d_g}$ represents the set of learnable parameters. d_g denotes the embedding dimension of the cell line feature after transformation.

Thereafter, we perform a linear transformation of the drug molecular fingerprint to obtain the initial features of drugs $X_f \in \mathbb{R}^{n \times d_f}$:

$$X_f = D_f \cdot W_f \tag{14}$$

where, $D_f \in \mathbb{R}^{n \times d_d}$ is the drug molecular fingerprint matrix, n represents the number of drugs, d_d is the length of the drug molecular fingerprint features. The term $W_f \in \mathbb{R}^{d_d \times d_f}$ denotes the set of learnable parameters for the transformation of drug features, d_f denotes the embedding dimension of the drug feature after transformation.

3.4 Low-Pass and High-Pass Hypergraph Convolutional Representation Learning

Hypergraphs can provide higher-order correlations between nodes. By applying low-pass and high-pass filters to hypergraphs, we can effectively extract both common and different features among hypergraph nodes.

Given a cell line hypergraph H_c, its normalized graph Laplacian operator is defined as $L_c = I_N - D^{-\frac{1}{2}} H_c D^{-\frac{1}{2}}$ and $D = diag\left(\sum_{i \neq j} H_{c(i,j)}\right)$. L_c summarizes the signal differences between neighbors and is regarded as a high-pass filter of the hypergraph H_c [14]. Hence, we obtained the cell line features $X_{ch}^{(k)}$ through the the high-pass convolution by Eqs. 15 and 16

$$Z_{ch}^{(k)} = L_c X_{ch}^{(k-1)} W_{ch}^{(k-1)} \tag{15}$$

$$X_{ch}^{(k)} = \alpha Z_{ch}^{(k)} + \beta \left(Z_{ch}^{(k)} \odot X_c^{(0)} \right) \tag{16}$$

where, $X_c^{(0)} = X_g$ is the initial feature of cell line obtained from Eq. 13. To avoid the over-smoothing effect of convolution, we add the dot product between the initial features of the node and the features after convolution. W_h denotes the learnable parameters for high-pass convolution. α and β are hyperparameters used to balance the weights of aggregation and residuals. k is the number of layers of the convolution.

$S_c = D^{-\frac{1}{2}} H_c D^{-\frac{1}{2}}$ computes the average of neighboring node features and is a low-pass filter of the hypergraph H_c[15]. We obtain the node features of the low-pass convolution $X_{cl}^{(k)}$ through Eq. 18:

$$Z_{cl}^{(k)} = S_c X_{cl}^{(k-1)} W_{cl}^{(k-1)} \tag{17}$$

$$X_{cl}^{(k)} = \alpha Z_{cl}^{(k)} + \beta \left(Z_{cl}^{(k)} \odot X_c^{(0)} \right) \tag{18}$$

where W_{cl} represents the learnable parameters for low-pass convolution.

Similarly, we can utilize Eqs. 16 and 18 to learn the hypergraph high-pass and low-pass features $X_{dh}^{(k)}$ and $X_{dl}^{(k)}$ of drugs on the drug hypergraph, respectively, where replacing L_c and S_c with L_d and S_d that are computed from the drug hypergraph H_d. $X_d^{(0)}$=X_f is the initial drug features obtained from Eq. 14.

3.5 Heterogeneous Graph Convolutional Representation Learning

We constructed a heterogeneous network with cell lines and drugs as nodes. The known cell line-drug associations are edges. We computed the heterogeneous graph convolutional features for cell lines, denoted as $Cell_f^{(k)}$, using Eq. 20:

$$Z_{cell}^{(k)} = \left(D_c^{\frac{1}{2}} A D_d^{\frac{1}{2}} \right) Drug_f^{(k-1)} W_{c1}^{(k-1)} + \left(D_c^{-1} + I_c \right) Cell_f^{(k-1)} W_{c2}^{(k-1)} \tag{19}$$

$$Cell_f^{(k)} = \alpha Z_{cell}^{(k)} + \beta \left(Z_{cell}^{(k)} \odot Cell_f^0 \right) \tag{20}$$

where, $Cell_f^0$ and $Drug_f^0$ are the initial features of cell lines and drugs, obtained by Eqs. 13 and 14, respectively. $D_{c(ij)} = \sum_j A_{ij} + 1$ and $D_{d(ij)} = \sum_i A_{ji} + 1$ represent the degree matrices of the known cell line-drug association matrix A. W_{c1} and W_{c2} represents the learnable parameters for the heterogeneous graph convolution of cell lines. Subsequently, we calculated the drug features from the heterogeneous graph convolution, denoted as $Drug_f^{(k)}$, through Eq. 22:

$$Z_{drug}^{(k)} = \left(D_d^{\frac{1}{2}} A^T D_c^{\frac{1}{2}} \right) Cell_f^{(k-1)} W_{d1}^{(k-1)} + \left(D_d^{-1} + I_d \right) Drug_f^{(k-1)} W_{d2}^{(k-1)} \tag{21}$$

$$Drug_f^{(k)} = \alpha Z_{drug}^{(k)} + \beta \left(Z_{drug}^{(k)} \odot Drug_f^0 \right) \tag{22}$$

where, W_{d1} and W_{d2} are the learnable parameters for the heterogeneous graph convolution of drugs.

3.6 Anticancer Drug Response Prediction

After learning features for cell lines and drugs from hypergraph and heterogeneous graph, we combine these features to obtain the final representations for cell lines $Cell_{final}$ and drugs $Drug_{final}$, through Eqs. 23 and 24, respectively:

$$Cell_{final} = ReLu\left(X_{ch}^{(k)} + X_{cl}^{(k)} + Cell_{f}^{(k)}\right) \tag{23}$$

$$Drug_{final} = ReLu\left(X_{dh}^{(k)} + X_{dl}^{(k)} + Drug_{f}^{(k)}\right) \tag{24}$$

where, *ReLu* represents the activation function.

Thereafter, we adopt Eq. 25 compute the linear correlation coefficient between the cell line's and the drug's final features:

$$Corr\left(h_i, h_j\right) = \frac{(h_i - \mu_i)\left(h_j - \mu_j\right)^T}{\sqrt{(h_i - \mu_i)(h_i - \mu_i)^T}\sqrt{\left(h_j - \mu_j\right)\left(h_j - \mu_j\right)^T}} \tag{25}$$

where $h_i \in Cell_{final}$ and $h_j \in Drug_{final}$ denote the features of the *ith* cell line and the *jth* drug, respectively. Finally, the cell line-drug association matrix is reconstructed as follows:

$$\hat{A} = \varphi\left(Corr\left(Cell_{final}, Drug_{final}\right)\right) \tag{26}$$

where, $\varphi(h) = \frac{1}{1+e^{-\gamma h}}$ represents the activation function, and the values in $\widehat{A}$ correspond to the predicted anticancer drug response labels. We employ the following loss function to constrain the model:

$$Loss\left(A, \hat{A}\right) = -\frac{1}{m \times n}\sum_{i,j} M_{ij}\left[A_{ij} ln\left(\hat{A}_{ij}\right) + \left(1 - A_{ij}\right) ln\left(1 - \hat{A}_{ij}\right)\right] \tag{27}$$

The matrix M denotes whether the cell line-drug association is present in the training set. If the relationship between cell line i and drug j is utilized for training, then $M_{ij} = 1$, otherwise, $M_{ij} = 0$.

We performed a grid search over the training set with five-fold cross-validation to identify the best hyperparameter configuration. Ultimately, the embedding dimensions for cell line gene expression and drug fingerprints were 2040. The Sigmoid scaling parameter was set to 8.7, the values of α and β were set at 0.85 and 0.15. k is 1. The learning rate was 0.001, weight decay was determined to be 1e−5, and the epoch count was 1000.

4 Experiments

4.1 Experimental Setup

To evaluate the effectiveness of HRLCDR, we compared it against six state-of-the-art baselines: DeepCDR [6], GraphCDR [8], DeepDSC [4], NIHGCN [9], DeepDSC, MOF-GCN [7] and TSGCNN [10].

To ensure a fair comparison, we used the same input data (gene expression, fingerprints, or molecular graphs) for the baseline methods as reported in their original research. We adopted the recommended parameter settings from their respective papers. We randomly selected 10% of the positive samples and the same number of negative samples to form an independent test set. The remaining data were used for cross-validation. We performed five five-fold cross-validation on the cross-validation set. This process was repeated five times and the average results were finally reported.

4.2 Randomly Zeroing Values Cross-Validation

Table 1 reports the experimental results for all algorithms on GDSC and CCLE. Our model, HRLCDR, outperformed all baseline methods in terms of classification prediction performance. In the GDSC dataset, our model achieved an AUC score of 87.68% and an AUPRC of 88.21% (0.14% and 0.20% improvement over the second-ranked method). In the CCLE dataset, our model attained an AUC score of 86.26% and an AUPRC of 86.29%. The model shows a 0.22% improvement in AUC over the second-best method, NIHGCN. Compared to models like DeepCDR and DeepDSC that don't focus on the known relationships between cell lines and drugs, methods like MOFGCN, GraphCDR, NIHGCN, TSGCNN, and especially HRLCDR all benefit from exploiting these known connections, which allows them to learn more effective representations for prediction. HRLCDR surpasses other graph convolution-based models (like NIHGCN and GraphCDR) because it incorporates hypergraph learning in addition to first-order associations. Hypergraphs allow HRLCDR to capture higher-order relationships between nodes. This advanced information aggregation leads to better cell line and drug feature representations, resulting in superior performance.

Table 1. Comparison of every method under randomly zeroing cross-validation on two datasets

Algorithm	GDSC		CCLE	
	AUC	AUPRC	AUC	AUPRC
DeepCDR	$0.8030 \pm 3 \times 10^{-3}$	$0.8064 \pm 3 \times 10^{-3}$	$0.8384 \pm 6 \times 10^{-5}$	$0.8292 \pm 2 \times 10^{-4}$
DeepDSC	$0.8083 \pm 7 \times 10^{-5}$	$0.8130 \pm 8 \times 10^{-5}$	$0.8543 \pm 5 \times 10^{-5}$	$0.8489 \pm 5 \times 10^{-5}$
MOFGCN	$0.8685 \pm 3 \times 10^{-6}$	$0.8726 \pm 3 \times 10^{-6}$	$0.8497 \pm 5 \times 10^{-5}$	$0.8514 \pm 7 \times 10^{-5}$
GraphCDR	$0.8176 \pm 6 \times 10^{-5}$	$0.8253 \pm 5 \times 10^{-5}$	$0.8396 \pm 1 \times 10^{-4}$	$0.8367 \pm 1 \times 10^{-4}$
NIHGCN	$0.8754 \pm 3 \times 10^{-6}$	$0.8801 \pm 3 \times 10^{-6}$	$0.8604 \pm 7 \times 10^{-5}$	$0.8623 \pm 6 \times 10^{-5}$
TSGCNN	$0.8706 \pm 3 \times 10^{-6}$	$0.8745 \pm 4 \times 10^{-6}$	$0.8524 \pm 7 \times 10^{-5}$	$0.8542 \pm 6 \times 10^{-5}$
HRLCDR	$\mathbf{0.8768 \pm 2 \times 10^{-6}}$	$\mathbf{0.8821 \pm 3 \times 10^{-6}}$	$\mathbf{0.8626 \pm 8 \times 10^{-5}}$	$\mathbf{0.8629 \pm 2 \times 10^{-4}}$

5 Conclusion

This article introduces HRLCDR, a novel drug response prediction model utilizing hypergraph representation learning. Compared to existing methods, HRLCDR employs low-pass and high-pass hypergraph convolutions to learn higher-order interaction features between cell lines and drugs. Meanwhile, the model learns features from a heterogeneous graph. Ultimately, the model reconstructs cell line-drug associations by integrating the two kinds of features learned from the hypergraph and heterogeneous graph. To assess HRLCDR's effectiveness, we compared its performance against state-of-the-art models. The results demonstrate that HRLCDR outperforms existing methods, achieving superior drug response prediction accuracy. Future work will focus on investigating more advanced hypergraph computation methods and developing effective drug molecular feature representations to enhance the performance of drug response prediction.

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References

1. Chan, H.S., Shan, H., Dahoun, T., Vogel, H., Yuan, S.: Advancing drug discovery via artificial intelligence. Trends Pharmacol. Sci. **40**, 592–604 (2019)
2. Ghandi, M., et al.: Next-generation characterization of the cancer cell line encyclopedia. Nature **569**, 503–508 (2019)
3. Yang, W., et al.: Genomics of Drug Sensitivity in Cancer (GDSC): a resource for therapeutic biomarker discovery in cancer cells. Nucleic Acids Res. **41**, D955–D961 (2012)
4. Li, M., et al.: DeepDSC: a deep learning method to predict drug sensitivity of cancer cell lines. IEEE/ACM Trans. Comput. Biol. Bioinf. **18**, 575–582 (2019)
5. Su, R., Liu, X., Wei, L., Zou, Q.: Deep-Resp-Forest: a deep forest model to predict anti-cancer drug response. Methods **166**, 91–102 (2019)
6. Liu, Q., Hu, Z., Jiang, R., Zhou, M.: DeepCDR: a hybrid graph convolutional network for predicting cancer drug response. Bioinformatics **36**, i911–i918 (2020)
7. Peng, W., Chen, T., Dai, W.: Predicting drug response based on multi-omics fusion and graph convolution. IEEE J. Biomed. Health Inform. **26**, 1384–1393 (2022)
8. Liu, X., Song, C., Huang, F., Fu, H., Xiao, W., Zhang, W.: GraphCDR: a graph neural network method with contrastive learning for cancer drug response prediction. Briefings in Bioinformatics 23, bbab457 (2022)
9. Peng, W., Liu, H., Dai, W., Yu, N., Wang, J.: Predicting cancer drug response using parallel heterogeneous graph convolutional networks with neighborhood interactions. Bioinformatics **38**, 4546–4553 (2022)
10. Peng, W., Chen, T., Liu, H., Dai, W., Yu, N., Lan, W.: Improving drug response prediction based on two-space graph convolution. Comput. Biol. Med. **158**, 106859 (2023)
11. Feng, Y., You, H., Zhang, Z., Ji, R., Gao, Y.: Hypergraph neural networks. In: Proceedings of the AAAI conference on artificial intelligence, pp. 3558–3565 (2019)
12. Bretto, A.: Hypergraph theory. An introduction. Mathematical Engineering, p. 1. Springer, Cham (2013). https://doi.org/10.1007/978-3-319-00080-0
13. Jin, S., et al.: A general hypergraph learning algorithm for drug multi-task predictions in micro-to-macro biomedical networks. PLoS Comput. Biol. **19**, e1011597 (2023)

14. Bruna, J., Zaremba, W., Szlam, A., LeCun, Y.: Spectral networks and locally connected networks on graphs. arXiv preprint arXiv:1312.6203 (2013)
15. Bojchevski, A., et al.: Scaling graph neural networks with approximate pagerank. In: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, pp. 2464–2473 (2020)